Isaiah Milkey

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Education

Arizona State University, Barrett Honors College Tempe, AZ
Bachelor of Science in Computer Science | GPA: 3.64 *May 2026*
Master of Science in Computer Science *Expected May 2027*
Relevant Courses: Foundations of Machine Learning, Deep Learning Structures, Artificial Intelligence, Natural Language Processing, Data Structures & Algorithms, Applied Linear Algebra, Data Visualization

Research Experience

Agentic AI Researcher Aug. 2025 – Present
LENS Lab, Arizona State University *Tempe, AZ*

- Contributed to a AI physics evaluation framework assessing video models (NVIDIA, Sora, Google Veo) using optical flow and object segmentation embeddings to detect anomalies and artifacts in AI Videos.
- Conducted rigorous error analysis and interpretability investigations to characterize failure modes in multimodal and LLM-based systems, informing architectural improvements and experiment design.
- Designed multi-agent architectures with RAG and graph-based memory to improve structured reasoning across complex, document-like benchmarks.

Perovskite Materials Researcher May 2025 – Present
Rolston Lab, Arizona State University *Tempe, AZ*

- Led a multimodal ML project combining LLMs and classical classifiers to detect material quality anomalies from spectroscopy data; applied prompt-tuning to LLaMA-2-7B, improving data interpretation accuracy by **8%**.
- Built an end-to-end scikit-learn pipeline for feature extraction and multi-model benchmarking (Logistic Regression, Random Forest, SVM, KNN), achieving up to **90% cross-validated accuracy** on limited labeled samples.
- Developed an LLM-based tool to extract and interpret structured features from photoluminescence data, enabling automated anomaly detection and replicable analysis of degradation behavior.

Projects

Intel Processor Defect Classification – Few-Shot Learning Mar. 2026 – Apr. 2026

- Designed a 2-stage cascade classifier for visual anomaly and defect detection on a DINOv2 ViT-Small backbone, decoupling binary defect detection from fine-grained classification on a severely imbalanced dataset (94.5% non-defect).
- Applied focal loss, tau-normalization, logit adjustment, and TTA-averaged class prototypes to boost balanced accuracy from **28% → 90.9%** and defect recall from ∼20% → 91.25%; achieved **100% recall on 6 of 8 defect classes**.
- Enabled few-shot generalization: new defect classes registered with **∼80% accuracy from a single labeled sample** and no retraining, demonstrating scalable detection under data-scarce real-world conditions.

CT Scan Battery Anomaly Detection Aug. 2025 – Dec. 2025

- Developed an unsupervised anomaly detection pipeline using PCA and K-Means to identify subtle manufacturing variations across **500 cylindrical lithium-ion battery CT scans**; resulted in a working paper.
- Reduced 14 CT-derived geometric features to 8 principal components retaining **96.1% of total variance**, enabling efficient and interpretable clustering of complex visual data at scale.
- Achieved **93.8% separation** of a known anomalous production batch (122/130 cells) with ARI of 0.31 against ground-truth batch labels; identified core geometry features as primary structural variation drivers.

MilkBot – Modular LLM Agent with Tool Use & RAG Aug. 2025 – Dec. 2025

- Built a modular LLM agent that classifies and routes unstructured queries through specialized reasoning and tool-augmented pipelines, integrating RAG over a LangChain vector DB with LLM-based verification.
- Doubled QA accuracy from **30% → 60%** on a multi-domain benchmark using self-consistency voting, retrieval, and domain-specific inference strategies.
- Designed automated batch evaluation infrastructure for systematic error analysis and iterative model improvement across diverse query types.

Additional Materials

Milkey, I. (December 2025). *Unsupervised Clustering and Dimensionality Reduction for Identifying Subtle Build Variation in Cylindrical Battery CT Scans.* Working Paper, School of Augmented Intelligence, Arizona State University. [link]

AgentsWorld Hackathon Winner - Audio Embedding | *Python, Voxel51, Embedding* Mar. 2026

- Built a Voxel51 plugin enabling interactive visualization and playback of audio datasets, with a backend that automatically partitions audio files into their most acoustically impactful segments for targeted analysis.
- Implemented audio embedding via the Voxel51 Brain to map segmented audio into vector space, enabling unsupervised clustering and anomaly detection across large unlabeled audio datasets.
- Won **Grand Prize** at the ASU × Voxel51 Hackathon competing against 20+ teams building agent-based and multimodal data tools; collaborated cross-functionally across ML, audio, and frontend (React/JS) workstreams.

Technical Skills

Languages: Python, C++, SQL, R
Frameworks: PyTorch, scikit-learn, Hugging Face, NumPy, Pandas, LangChain
Developer Tools: Git, Linux, VS Code, MATLAB
ML / AI: CNNs, Vision Transformers (ViT), LLMs, Multimodal Models, Anomaly Detection, Few-Shot Learning, RAG, Dimensionality Reduction, Clustering